Pie Charts

Exam code: 0607

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Updated on 20 January 2026

Pie charts

What is a pie chart?

- A **pie chart** is a circle divided into sectors which is used to **present data**
- The **sectors** represent different **categories**
  - They show the relative **proportions** of the categories
  - They do **not** show the actual **frequencies** of each category

How do I draw a pie chart?

- This is easiest explained through an example
- The following data shows the favourite colours of a class of students

Colour	Red	Purple	Blue	Green	Yellow	Orange
Students	11	4	9	3	2	1

- Write the frequencies as **fractions** of the **total number** of students, 30

Colour	Red	Purple	Blue	Green	Yellow	Orange
Students	11	4	9	3	2	1
Fractions	$\frac{11}{30}$	$\frac{4}{30}$	$\frac{9}{30}$	$\frac{3}{30}$	$\frac{2}{30}$	$\frac{1}{30}$

- Find the **angles** of the sectors by **multiplying each fraction** by **360°**
  - Then check all angles add up to 360°

Colour	Red	Purple	Blue	Green	Yellow	Orange
Students	11	4	9	3	2	1
Fractions	$\frac{11}{30}$	$\frac{4}{30}$	$\frac{9}{30}$	$\frac{3}{30}$	$\frac{2}{30}$	$\frac{1}{30}$

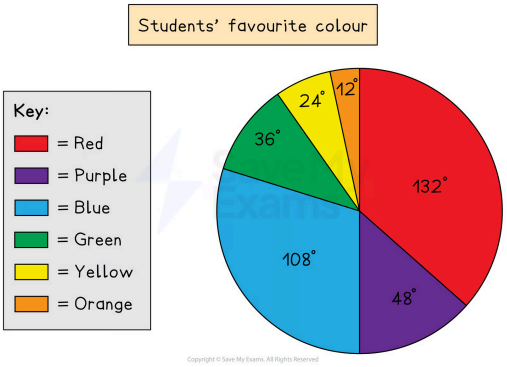
Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- Draw a vertical line from the circle's centre to the top, then use a **protractor** to mark off the first angle
  - Draw a line from the centre to this first mark
    - Then, from this line, mark off the next angle (and so on)



Example of a pie chart

How do I solve problems with pie charts?

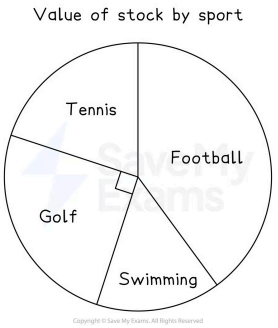
- Use the following facts
  - **angles** are **proportional** to the **frequencies** of each category
  - **360°** represents the **total frequency**
- For harder problems, it helps to work out
  - **what frequency** is represented **by 1°**
  - **what angle** is represented by **1 unit** of **frequency**
- For example, if a sector of 30° represents 15 people, then
  - 1° = 0.5 people (dividing by 30)
  - 2° = 1 person (multiplying by 2)
- These relationships can then be scaled **up** or **down** accordingly
  - If 1° = 0.5 people
    - then 360° = 180 people (multiplying by 360)

Examiner Tips and Tricks

If the pie chart says 'not to scale', then examiners want you to use ratio and proportion methods to answer the questions. Don't measure angles using a protractor!

Worked Example

The diagram is not to scale.



(a) Given that the shop stocks £12 000 of golf items, find the total value of the shop's stock.

Answer:

Find a relationship between an angle and a value

$$90° = £12\,000$$

The total value is represented by 360°
Multiply by 4 to get from 90° to 360°

$$360° = 4 \times 12\,000$$

Total value is £48 000

You can also do this question by finding 1° first

(b) If the angle on the pie chart for tennis is 72°, find the value of tennis items that are stocked by the shop.

Answer:

It is quickest to find the fraction $\frac{72}{360}$ of the total value, found in part (a)

$$\frac{72}{360} \times 48\,000 = 9\,600$$

The value of tennis items is £9 600

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